SULIT

SECP2523 DATABASE

SESSION 2024/2025 - SEMESTER 1

ALTERNATIVE ASSESSMENT REPORT: PHASE 1

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Year / Program	2 / SECPH
Section	01
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Case Study	WBL Project
System Name	Koperasi KADA Online System
Group Name	TECHNICRAB

SECTION A (SYSTEM'S OVERVIEW)		
1.1	**Overall Description**	
		<< Describe the system's overview. >>
1.2	**Project Weaknesses or Improvement**	
		<< List the project's weaknesses or improvement that can be made to the proposed system. Provide an explanation of the proposed module to overcome the weaknesses. >>
1.3	**Proposed Module**	
		<< Describe a new module to overcome the project's weaknesses. >>
SECTION B (DATABASE PLANNING AND DESIGN)		
2.1	**Conceptual ERD**	
		<< Draw a conceptual ERD to represent the proposed module. >>
2.2	**Logical ERD**	
		<<Provide logical ERD of your module in the proposed system.>>
2.3	**Data Dictionary for Logical ERD**	
		<< Provide the data dictionary based on the logical ERD in Subsection 2.2.>>
2.4	**Relational Database Schema**	
		<<Provide all relations database schema of your proposed module.>>
2.5	**Normalization**	
		<< Illustrate the normalization process up until BCNF to every relation listed in 2.5. At the end of this section, list all normalized relations achieved. >>

SECTION A (SYSTEM'S OVERVIEW)

1.1 Overall Description

Our objective is to help Koperasi Kakitangan KADA Kelantan Berhad in transitioning from their current manual system to a fully integrated digital platform. In this WBL project case study, the module I took in charge are the loan application process and the ability to view loan application status.

Currently, the cooperative handles loan application manually, which involve heavy paperwork, frequent mistakes, and delays in application review and approval. Members must physically visit the cooperative to submit their loan applications and repeatedly check to see if their applications have been approved. This method is not only time-consuming but also prone to data mismanagement and loss of important documents.

The proposed system introduces an online loan application feature where members can submit their applications digitally through our website system. Once submitted, the system automatically processes and stores the application data in a secure database, minimizing manual intervention and reducing errors. Members can easily access to real-time updates on the loan application status page, whether it is pending, approved, or rejected. Members can also watch monthly financial status through our system in the dashboard whenever they want to check on their savings or dividends. This eliminates the need for physical follow-ups and ensures members are promptly informed of any changes.

From the administrative perspective, the system streamlines the loan management process. Administrators can review and verify loan applications through a centralized platform, update application statuses, make financial adjustments for members and generate reports with just a few clicks. By automating repetitive tasks, the system allows staff to focus on critical decision-making and improve overall operational efficiency. The integrated reporting tools also enable administrators to track loan application trends and monitor performance metrics with ease.

The new system also feature a secure, centralized database to store and manage member information effectively. This will eliminate the need for members to repeatedly provide their personal details for every interaction. The database will also ensure easy access and retrieval of records, significantly reducing the time and effort required for administrative tasks. By digitizing the loan application process, the system not only improves data accuracy and security but also enhances communication between members and the admins.

In summary, the loan application module addresses the inefficiencies of the current manual system by providing a secure, efficient, and transparent digital solution. It simplifies the application process for members, reduces the administrative workload, and fosters better connectivity between stakeholders. This initiative represents a significant step toward modernizing the cooperative's operations and delivering greater value to its members.

Group member's name	Matric No.	Module
Cheryl Cheong Kah Voon	A23CS0060	Member module (financial status & manage profile)
Chau Ying Jia	A23CS0213	Member module (apply for loan & loan application status)
Neo Li Xin	A23CS0253	Admin module (generating reports)
Lau Yee Wen	A23CS0099	Employee Module (apply membership & membership application status)
Woo Cheng Shuan	A23CS0283	Admin module (membership and loan application approval)

1.2 Project Weaknesses or Improvement

The system for loan application process and status updating for member viewing brings numerous benefits but it still has several potential weaknesses that need to be addressed to maximize its effectiveness and usability.

The first potential issue is that this system is develop by us, the most beginner in developing a complete system, I could say that it is still a very simple and designed for ease use system. Many features are still to be under improvement and many other features still could be implemented for the smoothest and best user experience. Members might face occasional website lagging, performance issues or technical issue. These issues need to be addressed. We should optimize the system's performance and make regular testing, audits and page connection enhancements to make sure the system could run smoothly and no other technical issues to delay the loan application process. We can try to develop an automated error diagnostics feature to notify the administration for debugging once error is detected.

Another concern lies in the user interface design. Currently, the interface of our system could be improved, more neatly organized and maybe could be easier to navigate. Members might find it challenging to locate specific information when filling the loan application form or to complete actions or steps for example way to navigate to the view loan status page efficiently. A more informative and intuitive interface is necessary to improve user engagement and overall system usability. We should modify the interface focuses on simplicity, neatness and easy navigation to make it more user-friendly and accessible.

These are the weaknesses and improvements we can make to ensure a inclusive, reliable and secure system, ultimately fostering trust and satisfaction among all stakeholders.

1.3Proposed Module

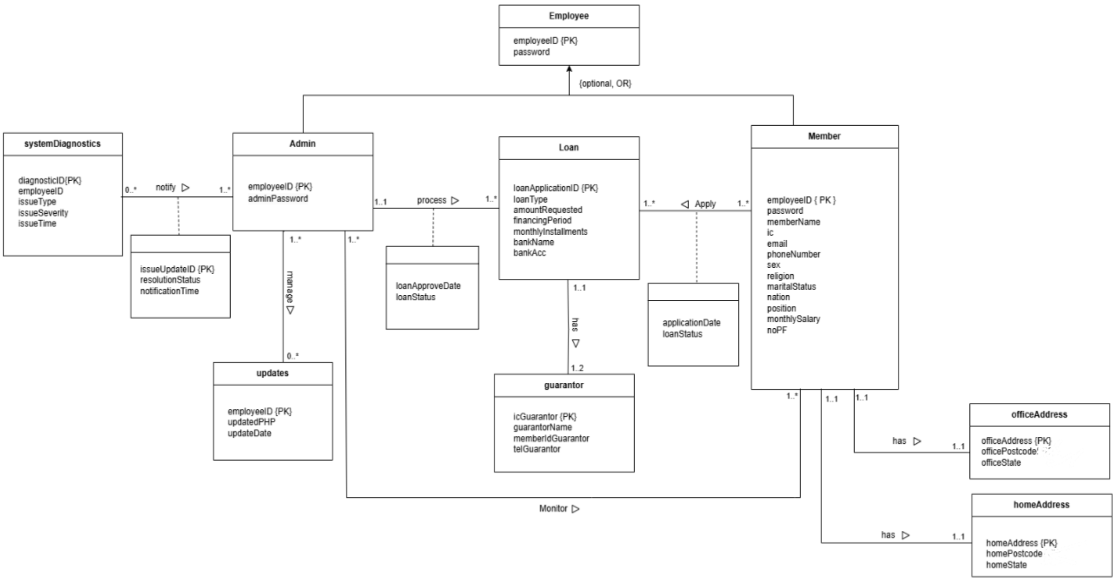
To address the weaknesses identified in current loan system module, I will propose the development module called "System improvement and Optimization Module". This module will enhance the system performance, improve user interface design, and provide robust solution to overcome technical and operational challenges. Here is a list of what can we do in this module to reach the expectations of our complete system:

1. Optimize codes configurations for better performance
2. Implement a load balancer to distribute traffic
3. Redesign the user interface to be more intuitive and neatly organized
4. Build in automated system diagnostics to automatic detect and notify of system issues
5. Strengthen security by implementing multi-layered authentication, encrypted communication protocols, and secure user access management.

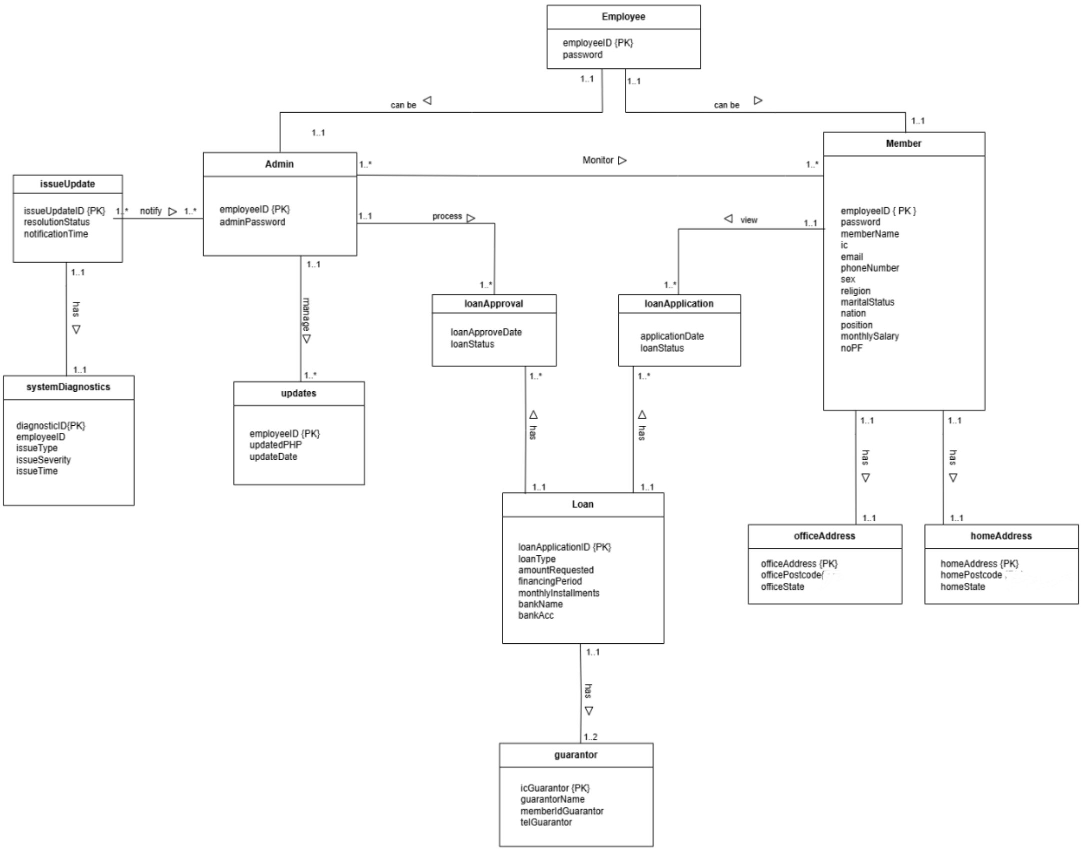
By incorporating this module, the loan system will overcome its current weaknesses to make a platform that meets the expectations of all stakeholder while preparing for future advancements too.

SECTION B (DATABASE PLANNING AND DESIGN)

2.1 Conceptual ERD



2.2 Logical ERD



2.3 Data Dictionary for Logical ERD

2.3.1 Identifying Entiti

Entity Name	Description	Aliases	Occurrence
Employee	Represents any person accessing the KADA system	New user, guest	Occurs when first-time user visit the website
Member	Represents registered users who have created KADA accounts	Members	Occurs when new user or guest register as a member
Admin	Represents the system admin who responsible for managing KADA system	Administrator, staff, board director	Occurs when system built and processing members' account activity
Loan	Represents loan applications submitted by members	Loan application record	Occurs when member apply for loan
guarantor	Represents information about members' loan guarantors	Members' loan application guarantor details	Occurs when member apply for loan
officeAddress	Stores members' office address	Members' personal information	Occurs when employer register as a member
homeAddress	Stores members' house address	Members' personal information	Occurs when employer register as a member
systemDiagnostics	Stores information when issues is detected	Admin	Occurs when detected there is an issue or error in the system
updates	Stores history of updated PHP files and dates of modification	Admin	Occurs when admin wants to modify codes or interface for better performances

2.3.2 Identifying Relationship Types

Entity Name	Multiplicity	Relationship	Entity Name	Multiplicity
Employee	1..1 1..1	can be can be	Admin member	1..1 1..1
Member	1..1 1..1 1..1	Has Has View	homeAddress officeAddress loanApplication	1..1 1..1 1..*
Admin	1..* 1..1 1..1	monitor process manage	Member loanApproval updates	1..* 1..* 1..*
issueUpdate	1..* 1..1	notify has	admin systemDiagnostics	1..* 1..1
Loan	1..1 1..1 1..1	has has has	guarantor loanApproval loanApplication	1..2 1..* 1..*

2.3.3 Attributes description

Entity Name	Attributes	Data Type & Length	Description	Nulls	Multi-valued	Example
Employee	employeeID(PK)	INT(4)	Unique ID for the employee	No	No	0001
	password	VARCHAR (20)	User's account password	No	No	P12345W
Member	employeeID (PK)	INT(4)	Unique ID for member	No	No	0001
	password	VARCHAR (30)	Password for member login	No	No	P12345W
	memberName	VARCHAR (80)	Name of the member	No	No	Ali Abu
	ic	CHAR (12)	Identification number of the member	No	No	040101081234
	email	VARCHAR (30)	Member's email address	No	No	member@gmail.com
	phoneNumber	VARCHAR (15)	Contact number of the member	No	No	0123456789
	sex	CHAR (1)	Gender of the member (M/F)	No	No	Female
	religion	VARCHAR (30)	Religion of the member	No	No	Buddha
	maritalStatus	VARCHAR (15)	Marital status	No	No	Married
	nation	VARCHAR (30)	Nationality of the member	No	No	Malaysia
	position	VARCHAR (30)	Working position of the member	No	No	Data Engineer

	monthlySalary	INT (6)	Member's monthly salary	No	No	3000
	noPF	INT(4)	Member no PF	No	No	5678
Admin	employeeID (PK)	INT(4)	Unique ID for admin	No	No	4567
	adminPassword	VARCHAR (20)	Password of the admin	No	No	noahlim
Loan	loanType	VARCHAR(30)	Type of loan	No	No	AL_BAI
	loanApplicationID (PK)	VARCHAR (20)	Unique ID for loan application	No	AUTO-INCREMENT	0003
	amountRequested	INT (7)	Amount requested for the loan	No	No	3000
	financingPeriod	INT(2)	Loan repayment period in months	No	No	5
	monthlyInstallments	INT(7)	Monthly repayment amount	No	No	650
	bankName	VARCHAR (50)	Bank name for loan transfer	No	No	RHB
	bankAccount	VARCHAR (20)	Bank account number	No	No	01234567890
guarantor	GuarantorName	VARCHAR (80)	Name of the guarantor	No	No	Evelyn Lee
	IcGuarantor (PK)	INT (12)	Identification number of the member	No	No	040101081234
	memberIdGuartor	INT (4)	Unique ID for member's guarantor	No	No	8888
	telGuarantor	VARCHAR (15)	Contact number of the member' guarantor	No	No	0123456789
officeAddress	officeAddress	VARCHAR (100)	Address of member's office	No	No	3A, Jalan Permai
	officePostcode	INT(6)	Postcode of office address	No	No	31400
	officeState	VARCHAR(30)	Office located state	No	No	Perak
homeAddress	homeAddress	VARCHAR (100)	Address of member's house	No	No	36, Jalan Indah Sakti
	homePostcode	INT(6)	Postcode of home address	No	No	31400

	officeState	VARCHAR (30)	Office located state	No	No	Perak
issueUpdate	issueUpdateID	INT (10)	Unique ID for each update	No	AUTO-INCREMENT	00000 00333
	resolutionStatus	VARCHAR (10)	Status on resolving errors	No	No	pending
	notificationTime	TIMESTAMP	Time when issues id detected and notification is sent to admin	No	No	2025-01-01 00:00:01
systemDiagnostic	diagnosticID(PK)	INT (10)	Unique ID for each diagnostic	No	AUTO-INCREMENT	00000 01117
	employeeID	INT (4)	Unique ID for employee who diagnose the issue	No	No	0004
	issueType	INT (20)	Type of issue	Yes	No	Server down
	issueSeverity	VARCHAR(20)	Severity of that issue	No	No	low
	issueTime	TIMESTAMP	Time when issue appear	No	No	2025-01-01 00:00:01
updates	employeeID	INT (4)	Unique ID for employee who diagnose the issue	No	No	0135
	updatedPHP	VARCHAR(40)	Name of PHP file of the file that make changes	No	No	Permohonanloan.php
	updateDate	TIMESTAMP	Date and time when admin made an update	No	No	2025-01-01 00:00:01
loanApproval	loanApprovalDate	TIMESTAMP	Date and time when loan is approve or decline	No	No	2025-01-01 00:00:01
	loanStatus	VARCHAR (10)	Status on loan application	No	No	rejected
loanApplication	applicationDate	TIMESTAMP	Date and time when member apply for loan	No	No	2025-01-01 00:00:01
	loanStatus	VARCHAR (10)	Status on loan application	No	No	rejected

2.4 Relational Database Schema

EMPLOYEE (employerID, password)
    PK: employeeID

ADMIN (employeeID, adminPassword)
    PK: employeeID
    FK: employeeID references EMPLOYEE (employeeID)

MEMBER(employeeID, password, memberName, ic, email, phoneNumber, sex, religion, maritalStatus, nation, position, monthlySalary, noPF)
    PK: employeeID
    FK: employeeID references EMPLOYEE (employeeID)

UPDATES (employeeID, updatePHP, updateDate)
    PK: employeeID
    FK: employeeID references ADMIN (employeeID)

ISSUEUPDATE (issueUpdateID, resolutionStatus, notificationtime, employeeID)
    PK: issueUpdateID
    FK: employeeID references ADMIN (employeeID)

SYSTEMDIAGNOSTICS (diagnosticID, employeeID, issueType, issueSeverity, issueTime, issueUpdateID)
    PK: diagnosticID
    FK: employeeID references ADMIN (employeeID)
    FK: issueUpdateID references ISSUEUPDATE (issueUpdateID)

LOAN(loanApplicationID, loanType, amountRequested, financingPeriod, monthlyInstallments, bankName, bankAcc)
PK: loanApplicationID

LOANAPPROVAL(employeeID, loanApplicationID, loanApproveDate, loanStatus)
    PK: employeeID, loanApplicationID
    FK: employeeID references ADMIN(employeeID)
    FK: loanApplicationID references LOAN(loanApplicationID)

LOANAPPLICATION(employeeID, loanApplicationID, applicationDate, loanStatus)
    PK: employeeID, loanApplicationID
    FK: employeeID references MEMBER(employeeID)
    FK: loanApplicationID references LOAN(loanApplicationID)

GUARANTOR (icGuarantor, loanApplicationID, guarantorName, memberIdGuarantor, telGuarantor)
    PK: icGuarantor, loanApplicationID
    FK: loanApplicationID references LOAN (loanApplicationID)

OFFICEADDRESS (officeAddress, officePostcode, officeState)
    PK: officeAddress, officePostcode

HOMEADDRESS(homeAddress, homePostcode, homeState)
    PK: homeAddress, homePostcode

2.5 Normalization

1. **employee – admin**

   1NF:

   Employee_Admin (employeeID, password, adminPassword)

   PK: employeeID

   2NF:

   Employee (employeeID, password)

   PK: employeeID

   Admin (employeeID, adminPassword)

   PK: employeeID

   FK: employeeID references Employee(employeeID)

   3NF:

   No changes, already in 3NF.

2. **employee – member**

   1NF:

   Employee_Member(employeeID, password, memberName, ic, email, phoneNumber, sex, religion, maritalStatus, nation, position, monthlySalary, noPF)

   PK: employeeID

   2NF:

   Employee (employeeID, password, position, monthlySalary)

   PK: employeeID

   Member (employeeID, memberName, ic, email, phoneNumber, sex, religion, maritalStatus, nation, noPF)

   PK: employeeID

   Employee_Member(employeeID)

   PK: employeeID
   FK: employeeID references Employee(employeeID)
   FK: employeeID references Member(employeeID)

   3NF:

   Does not exist.

3. admin – member

1NF:

Admin_member(employeeID, adminPassword, password, memberName, ic, email, phoneNumber, sex, religion, maritalStatus, nation, position, monthlySalary, noPF)

PK: employeeID

2NF:

Admin (employeeID, adminPassword)

PK: employeeID

Member (employeeID, password, memberName, ic, email, phoneNumber, sex, religion, maritalStatus, nation, position, monthlySalary, noPF)

PK: employeeID

Admin_member (employeeID)

PK: employeeID

FK: employeeID references Admin(employeeID)
FK: employeeID references Member(employeeID)

3NF:

Does not exist.

4. admin – issueUpdate

1NF:

Admin_issueUpdate (employeeID, issueUpdateID, adminPassword, resolutionStatus, notificationTime)

PK: employeeID, issueUpdateID

2NF:

Admin(employeeID, adminPassword)

PK: employeeID

IssueUpdate(issueUpdateID, resolutionStatus, notificationTime)

PK: issueUpdateID

Admin_issueUpdate (employeeID, issueUpdateID, adminPassword, resolutionStatus, notificationTime)

PK: employeeID, issueUpdateID
FK: employeeID references Admin (employeeID)
FK: issueUpdateID references issueUpdate (issueUpdateID)

3NF:

Does not exist.

5. issueUpdate – systemDiagnostics

1NF:

Issue_SystemDiagnostics (diagnosticID, issueUpdateID, employeeID, resolutionStatus, issueType, issueSeverity, issueTime, notificationTime)
PK: diagnosticID, issueUpdateID

2NF:

IssueUpdate(issueUpdateID, resolutionStatus, notificationTime)
PK: issueUpdateID
FK: employeeID references Admin(employeeID)

SystemDiagnostics(diagnosticID, employeeID, issueType, issueSeverity, issueTime)

PK: diagnosticID
FK: employeeID references IssueUpdate(employeeID)

Issue_SystemDiagnostics (diagnosticID, issueUpdateID, employeeID, resolutionStatus, issueType, issueSeverity, issueTime, notificationTime)
PK: diagnosticID, issueUpdateID
FK: diagnosticID references issueUpdate (diagnosticID)
FK: IssueUpdateID references issueUpdate (IssueUpdateID)
FK: employeeID references Admin(employeeID)

3NF:

Does not exist.

6. admin – updates

1NF:

Admin_Updates(employeeID, adminPassword, updatePHP, updateDate)
PK: employeeID

2NF:

Admin(employeeID, adminPassword)
PK: employeeID

Updates(employeeID, updatePHP, updateDate)
PK: employeeID
FK: employeeID references Admin(employeeID)

3NF:

Does not exist.

7. **admin – loanApproval**

1NF:

Admin_Updates(employeeID, adminPassword, loanApproveDate, loanStatus)
PK: employeeID

2NF:

Admin (employeeID, adminPassword)
PK: employeeID

LoanApproval (employeeID, loanApproveDate, loanStatus)
PK: employeeID,
FK: employeeID references Admin(employeeID

3NF:
Does not exist.

8. **loanApproval – loan**

1NF:

LoanApproval_Loan(employeeID, loanApplicationID, loanApproveDate, loanStatus, loanType, amountRequested, financingPeriod, monthlyInstallments, bankName, bankAcc)

PK: employeeID, loanApplicationID

2NF:

Loan(loanApplicationID, loanType, amountRequested, financingPeriod, monthlyInstallments, bankName, bankAcc)
PK: loanApplicationID

LoanApproval(employeeID, loanApplicationID)
PK: employeeID, loanApplicationID
FK: loanApplicationID references Loan(loanApplicationID)

3NF:

Does not exist.

9. loanApplication – loan

1NF:

LoanApplication_Loan(employeeID, loanApplicationID, applicationDate, loanStatus, loanType, financingPeriod, monthlyInstallments, bankName, bankAcc)
PK: employeeID, loanApplicationID

2NF:

Loan (loanApplicationID, loanType, financingPeriod, monthlyInstallments, bankName, bankAcc)
PK: loanApplicationID

LoanApplication(employeeID, loanApplicationID, applicationDate, loanStatus)
PK: employeeID, loanApplicationID
FK: employeeID references Member (employeeID)
FK: loanApplicationID references Loan(loanApplicationID)

3NF:

Does not exist.

10. loan – guarantor

1NF:

loan_guarantor(loanApplicationID, icGuarantor, loanType, amountRequested, financingPeriod, monthlyInstallments, bankName, bankAcc, guarantorName, memberIdGuarantor, telGuarantor)

PK: loanApplicationID, icGuarantor

2NF:

Guarantor (icGuarantor, guarantorName, memberIDGuarantor, telGuarantor)
PK: icGuarantor

loan_guarantor (loanApplicationID, icGuarantor, amountRequested, financingPeriod, monthlyInstallments, bankName, bankAccount)

PK: loanApplicationID, icGuarantor

FK: icGuarantor references guarantor (icGuarantor)

3NF:

Does not exist.

11. member – loanApplication

1NF:

member_loanApplication (employeeID, loanApplicationID, password, memberName, ic, email, phoneNumber, sex, religion, maritalStatus, nation, position, monthlySalary, noPF, applicationDate, loanStatus)
PK: employeeID, loanApplicationID

2NF:

Member (employeeID, password, memberName, ic, email, phoneNumber, sex, religion, maritalStatus, nation, position, monthlySalary, noPF)
PK: employeeID

loanApplication (loanApplicationID, employeeID, applicationDate, loanStatus)

PK: loanApplicationID, employeeID
FK: employeeID references member (employeeID)
FK: loanApplicationID references loan (loanApplicationID)

3NF:

Does not exist.

12. member – officeAddress

1NF:

Member_OfficeAddress (employeeID, officeAddress, officePostcode, officeState)
PK: employeeID, officeAddress

2NF:

Member (employeeID, password, memberName, ic, email, phoneNumber, sex, religion, maritalStatus, nation, position, monthlySalary, noPF)
PK: employeeID

OfficeAddress (officeAddress, officePostcode, officeState, employeeID)
PK: officeAddress
FK: employeeID references Member(employeeID)

3NF:

Does not exist.

13. member – homeAddress

1NF:

Member_homeAddress (employeeID, homeAddress, homePostcode, homeState)
PK: employeeID, homeAddress

2NF:

Member (employeeID, password, memberName, ic, email, phoneNumber, sex, religion, maritalStatus, nation, position, monthlySalary, noPF)
PK: employeeID

HomeAddress (homeAddress, homePostcode, homeState, memberID)
PK: homeAddress
FK: memberID references Member(memberID)

3NF:

Does not exist.